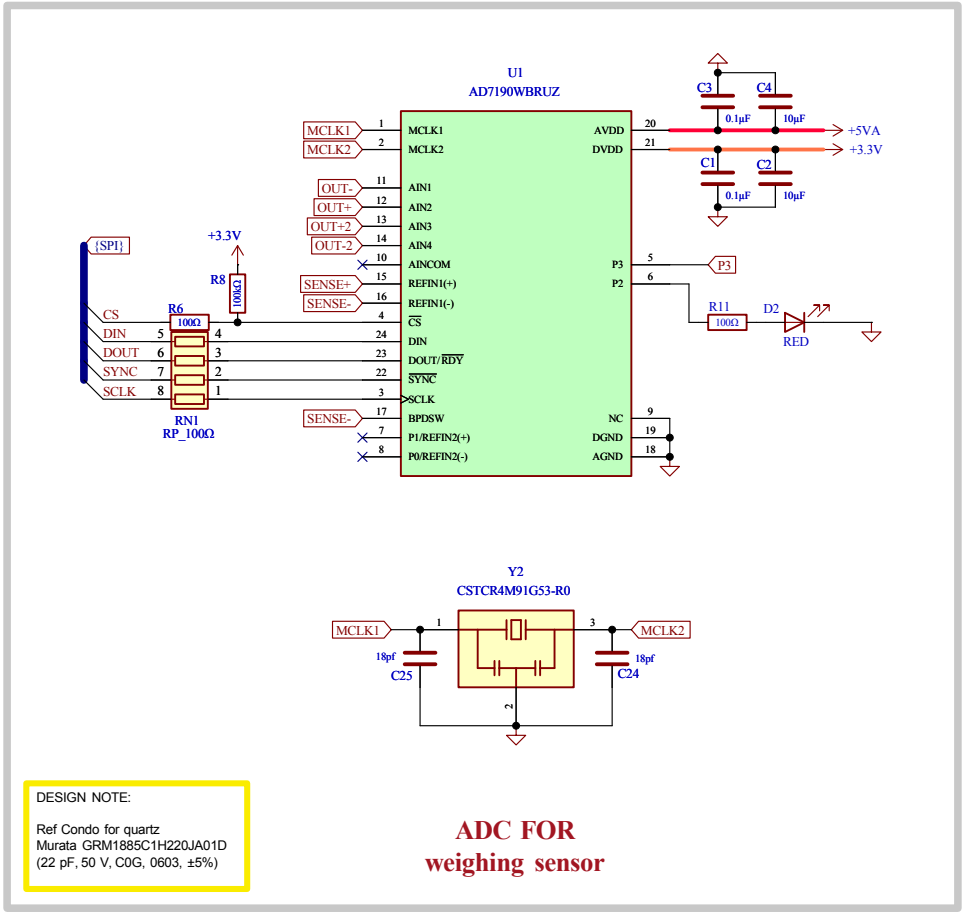
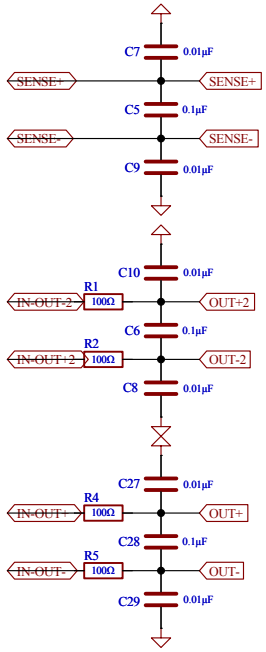
[4] AD7190 - Weight Sensor



DESIGN NOTE:

Ref Condo for quartz
Murata GRM1885C1H220JA01D
(22 pF, 50 V, C0G, 0603, ±5%)

ADC FOR weighing sensor



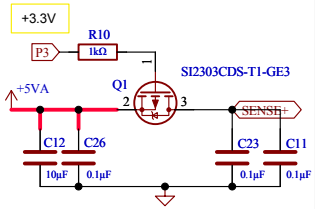
Calcule de filtre : fc = 1 / (2π × R × C) = X Hz

DESIGN NOTE:

Ref Condo for Filtre
Murata GRM1885C1H103JA01D (10 nF, 50 V, C0G, 0603, ±5%)
KEMET C0603R104K5RACTU (100 nF, 50 V, X7R, 0603, ±10%)

Ref Resistor for Filtre
TNPW0603100RBEEA (100 Ω, 0.1 W, thin film, 0603, ±0.1%, 25 ppm/°C)

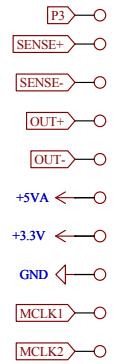
FILTRE



DESIGN NOTE:

IC1 is a P-MOS high-side switch used to enable/disable the load cell excitation.
must disabled during calibration

DRIVE GATE



TEST POINT

About

The AD7190 is a low noise, complete analog front end for high precision measurement applications. It contains a low noise, 24-bit sigma-delta (∑-Δ) analog-to-digital converter (ADC).
The on-chip low noise gain stage means that signals of small amplitude can be interfaced directly to the ADC.
The device can be configured to have two differential inputs or four pseudo differential inputs. The on-chip channel sequencer allows several channels to be enabled, and the AD7190 sequentially converts on each enabled channel. This simplifies communication with the part. The on-chip 4.92 MHz clock can be used as the clock source to the ADC or, alternatively, an external clock or crystal can be used. The output data rate from the part can be varied from 4.7 Hz to 4.8 kHz.
The device has two digital filter options. The choice of filter affects the rms noise/noise-free resolution at the programmed output data rate, the settling time, and the 50 Hz/60 Hz rejection.

Comments: AD7190 / CSTCR4M91G53-R0 / AD7190_TEST_BOARD CN	Company: OPEN THRUST LAB		Variant:	Git Hash: 349b121
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
Sheet Title: AD7190 - Weight Sensor	File Name: AD7190 - Weight Sensor.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Project Architecture/AD7190_Weight_Sensor/		Reviewer:	Size: A4	Sheet: 4 of 10